THE ASSAY

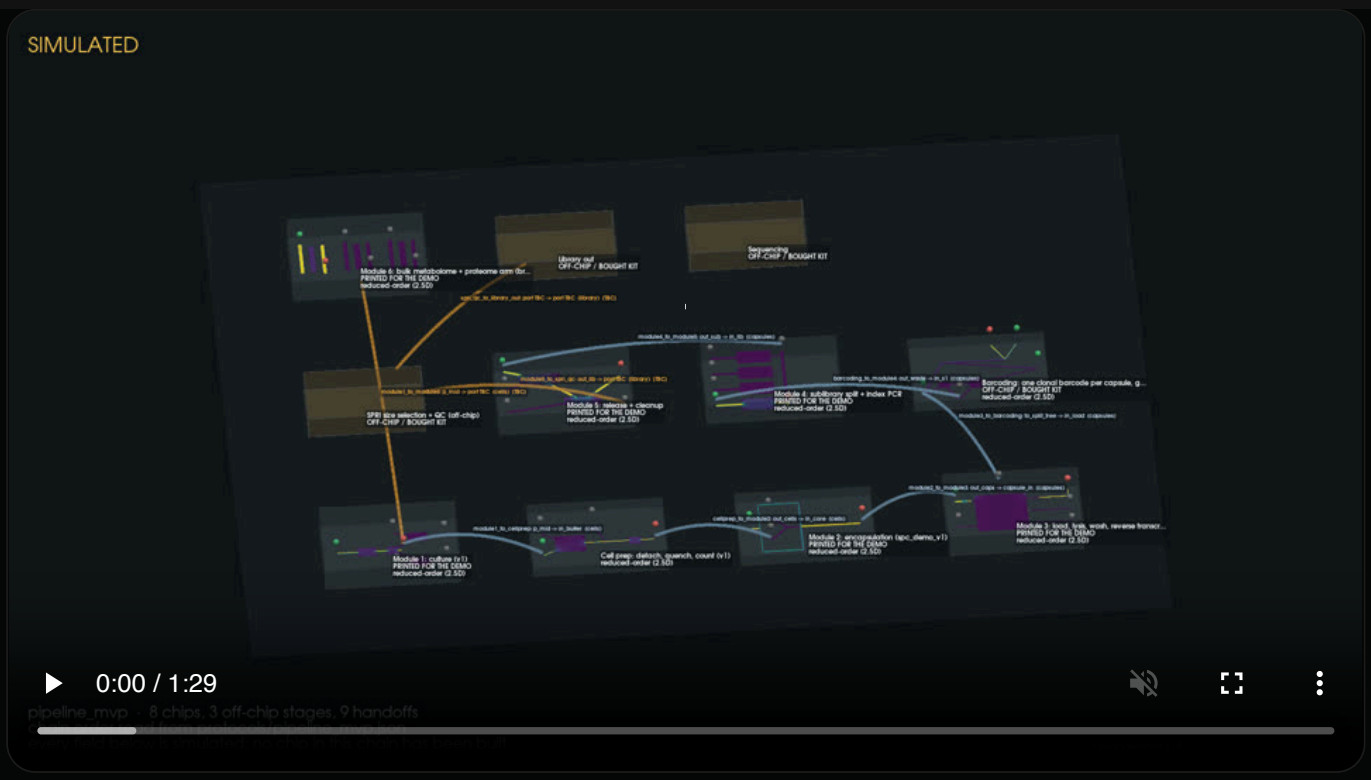
A capsule processor: small printed chips that grow three kinds of human brain cell side by side, then seal every cell in its own capsule so each one can be read on its own.

This site shows that assay stage by stage, runs the software that designs chips like these from a written lab protocol, and shows what its flow simulation found.

The demo

Three different things, each labelled: two photographs of a real printed chip running food dye, a simulation of flow through the whole chain, and a walkthrough of the digital twin. No cells, no biology has run yet.

SIMULATED FLOW
REDUCED-ORDER (2.5D)



Simulated flow, whole chain, 89 s. All eight chips on a bench in chain order with the nine handoff tubes, each chip's flow field and streamlines from its feed ports. The printed chips and the off-chip stages are badged as such. **Every chip here comes from the fast reduced-order (2.5D) solver.** The full 3D OpenFOAM cases exist but their fields are not converted yet, so nothing here is a 3D solve.

REAL HARDWARE · PHOTOS, NOT VIDEO · FOOD DYE, NOT CELLS

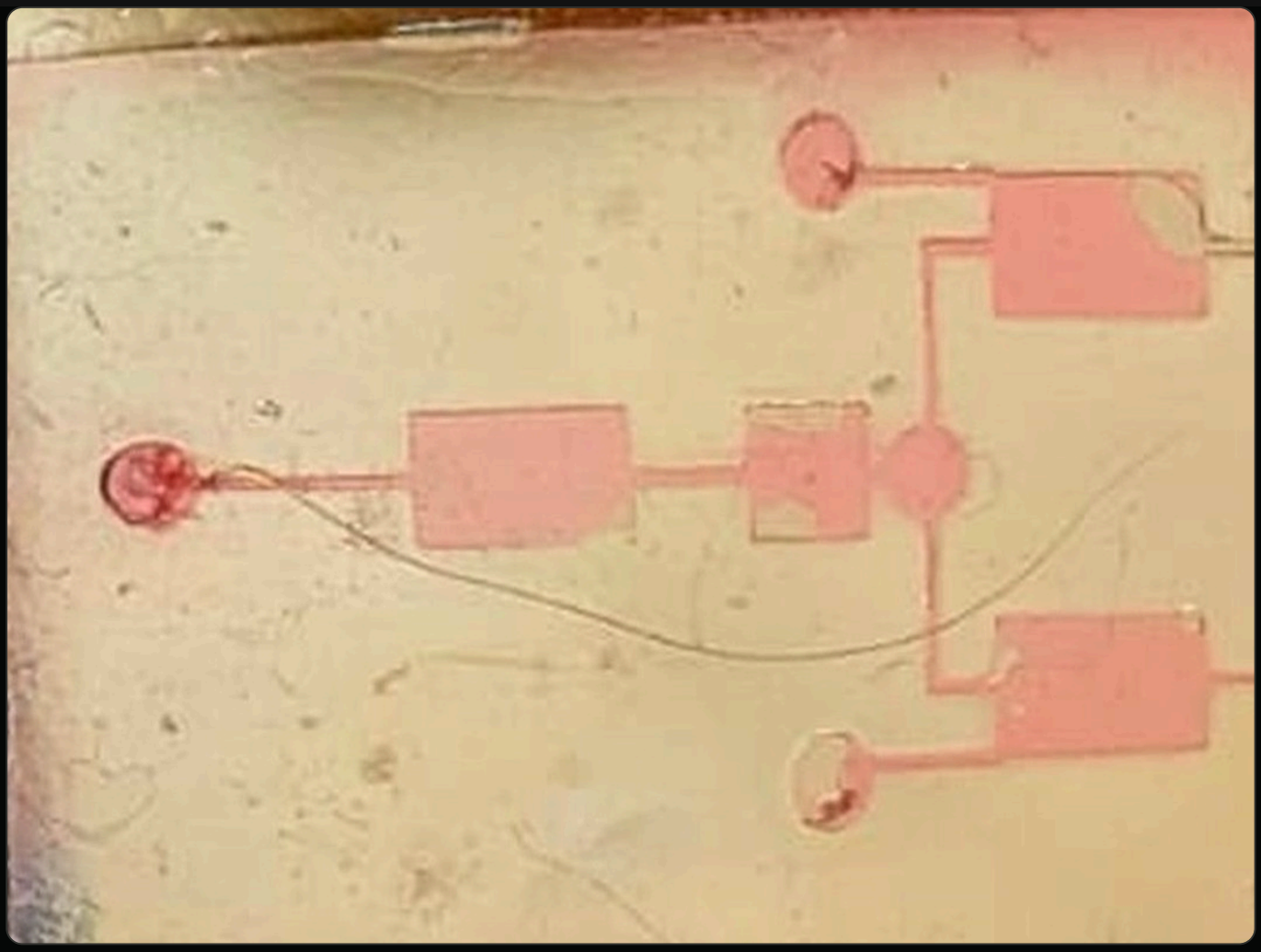


Photo: the printed Module 1 chip flooded with pink food dye.

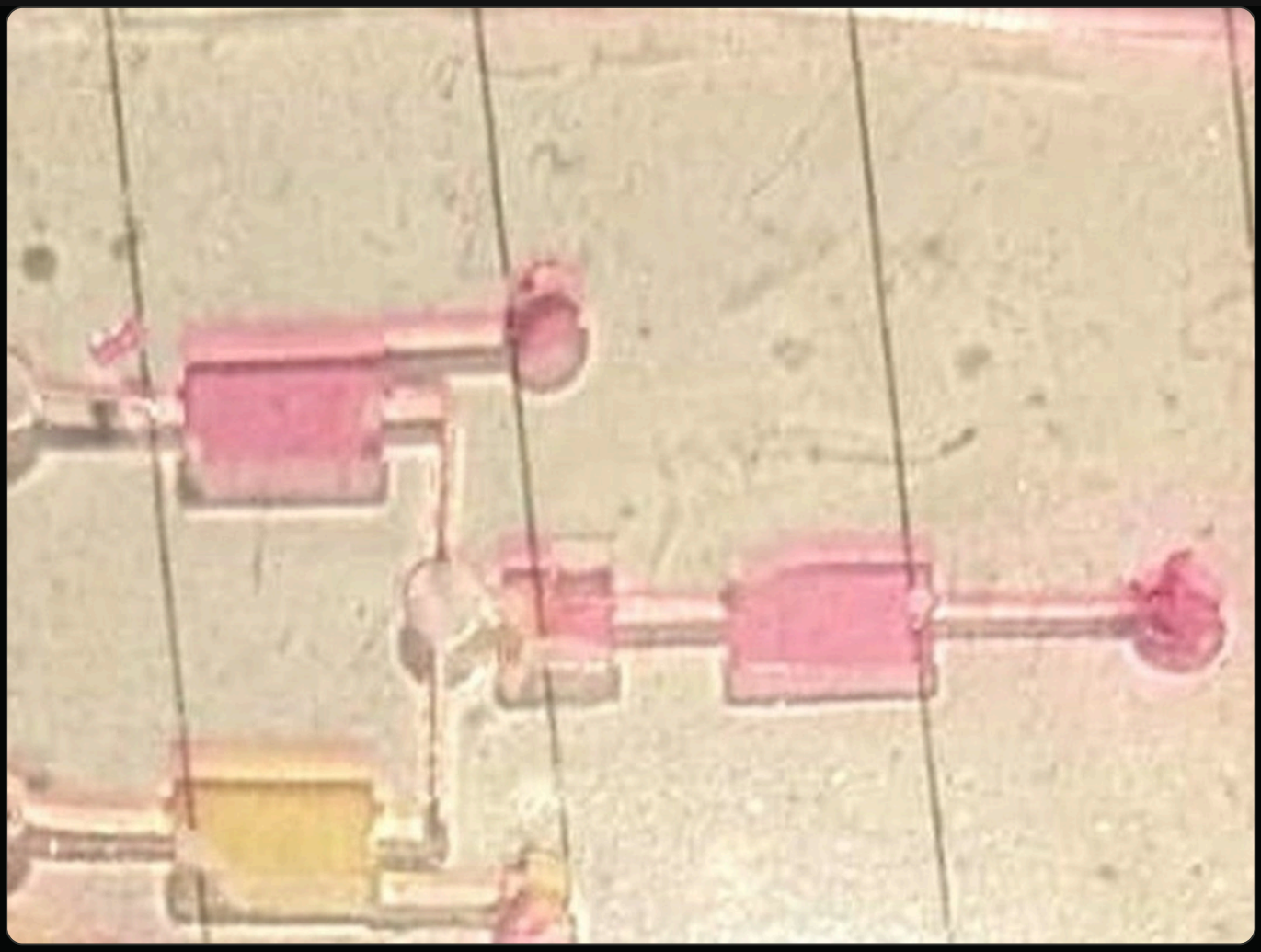


Photo: pink food dye in one chamber, yellow in the chamber next to it, held apart.

The team's own photographs of the printed Module 1 chip, from their presentation. There is no bench video of the dye running yet.

What the two bench photos show

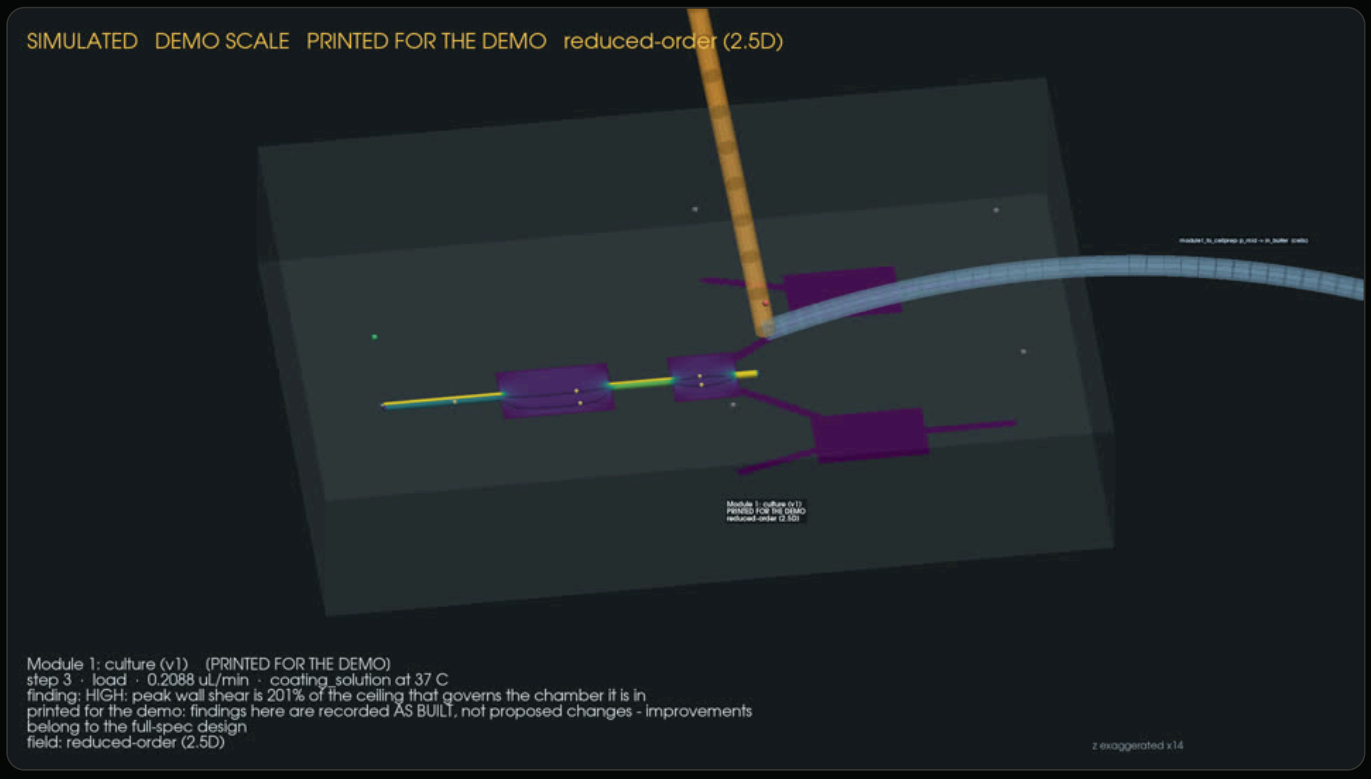
- A Module 1 chip was printed: these are photographs of a physical object.
- In the team's account of the first photo, dye entered at one inlet and reached every channel, chamber and port.
- In the second, two dyes sit in neighbouring chambers, which the team reads as the walls holding. That is the baseline every later perturbation is measured against.

What it does not show

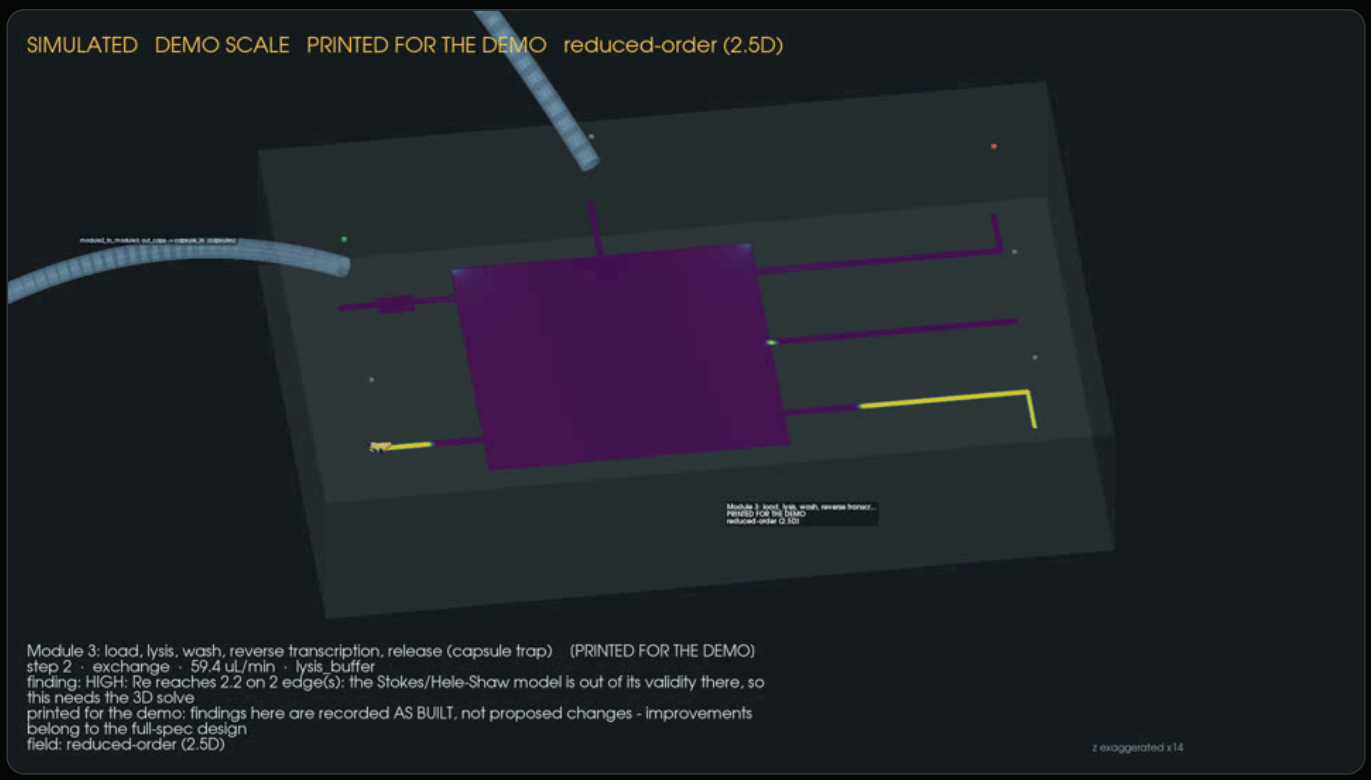
- Any cell biology. No cells have been in these chips.
- Anything about the other five chips: only Module 1 is photographed.
- Any thermal step, such as heating or PCR cycling.
- That the real reagents are compatible with the printed material.

- That capsules are retained in the chambers.

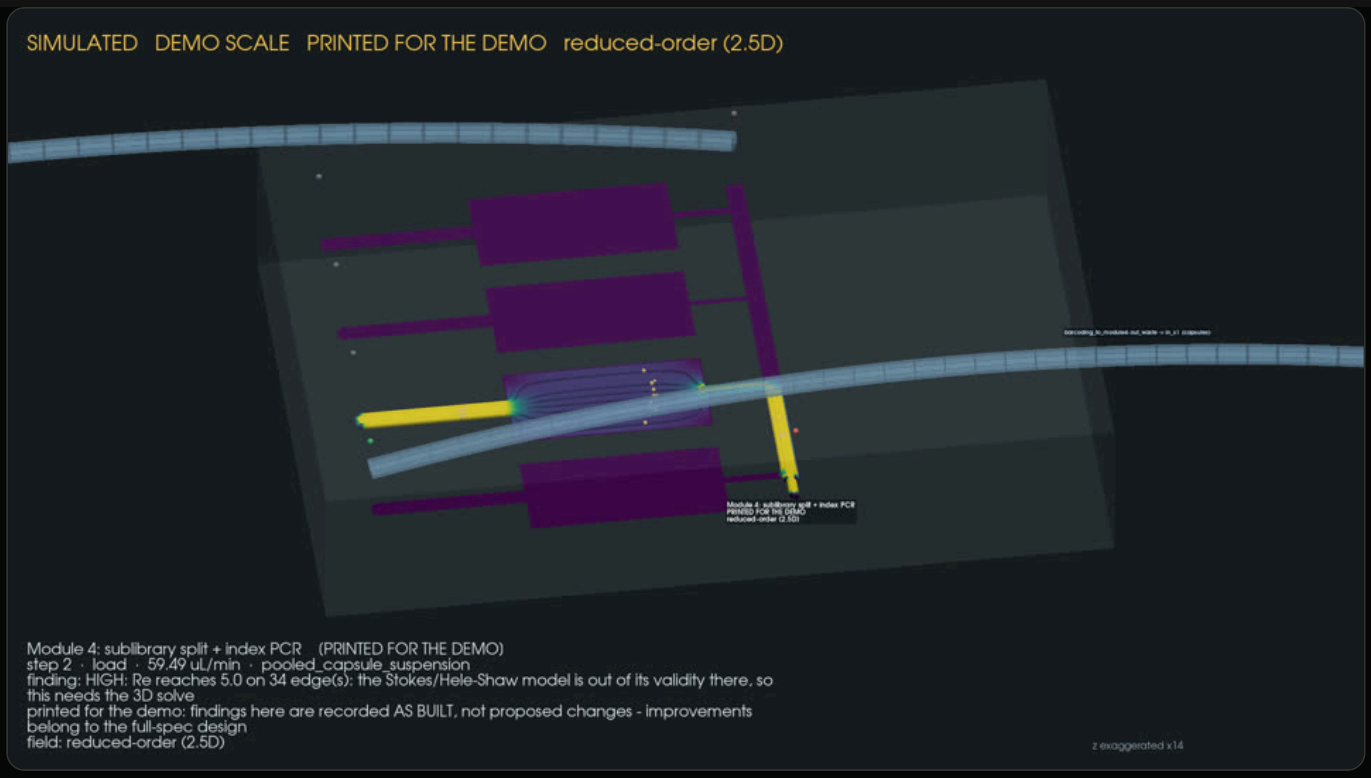
MORE SIMULATION



Module 1: simulated flow field and streamlines from its feed ports.

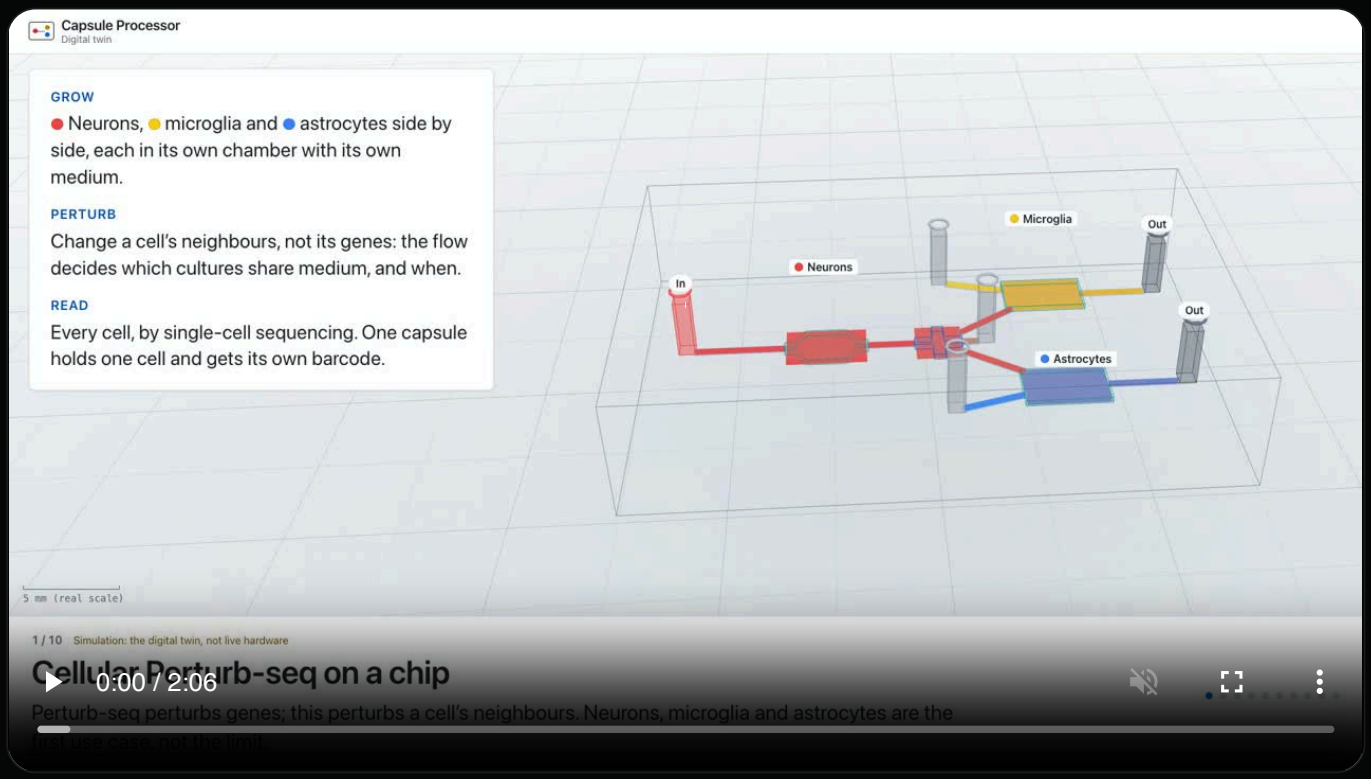


Module 3: capsule trap during the lysis-buffer exchange, flagged for the 3D solve.



Module 4: the sublibrary split, flagged where the reduced-order model runs out of validity.

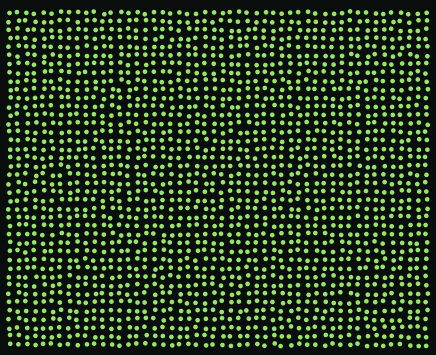
DIGITAL TWIN – SIMULATION, NOT LIVE HARDWARE



Digital twin walkthrough, 2 min 7 s. A screen recording of the browser digital twin: ten slides titled “Cellular Perturb-seq on a chip”, showing neurons, microglia and astrocytes in their own chambers and the chain around them. Its own footer reads “Simulation: the digital twin, not live hardware”. No physical chip appears in it.

How much one run holds

Approximate design figures for the full-size device. Not measurements.



≈ 2,000

cells per condition

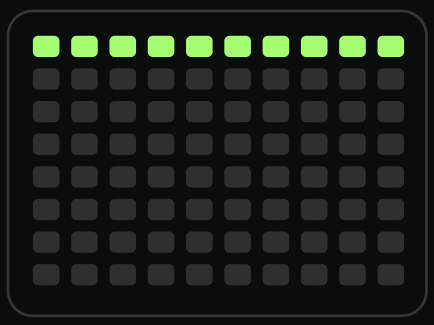
Each dot is one cell.



≈ 10

conditions per well

Each square is one condition of about 2,000 cells, in one well.



≈ 80

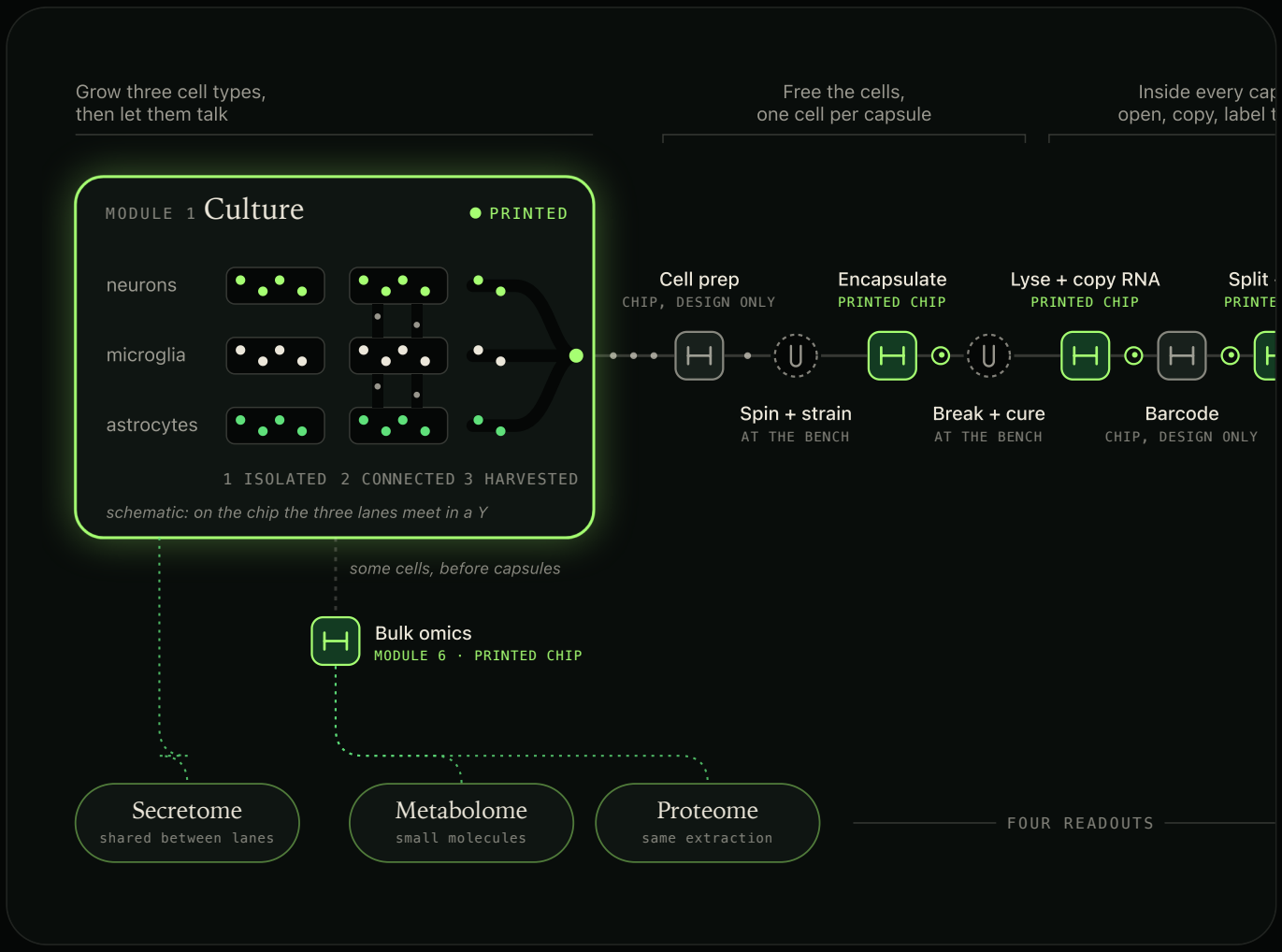
conditions per commercial chip

Each square is one condition. The bright ten are about one well's worth.

The chain, stage by stage

Follow one sample from left to right. Click a stage, or tab to it and press Enter, to see what it does.

13 stages: 8 on a chip, 6 of them printed for the demo, and the rest done off-chip.



The drawing is wider than this window: scroll it sideways to follow the whole chain.

a chip we printed · a chip, designed but not printed · done at the bench or on a bought kit

not built yet · loose cells travelling · capsules, one cell in each · DNA library

Each capsule is the reason this is single-cell: one cell per capsule, and later one barcode per capsule. Every readout at the bottom comes from a different point in the chain.

STAGE 1 OF 13 · PRINTED CHIP

Module 1: culture (v1)

PRINTED FOR THE DEMO · DEMO SCALE · FINAL

Grows the three cell types, each made from stem cells, in three separate lanes. The lanes are kept sealed while each type matures, then connected so the cells can signal to each other through the shared liquid, and finally all three are flushed out together through the middle junction.

ARRIVES

Nothing: the chain starts here.

Culture

LEAVES

cells to Cell prep

One counted suspension leaves j_mid through p_mid.

cells to Bulk omics

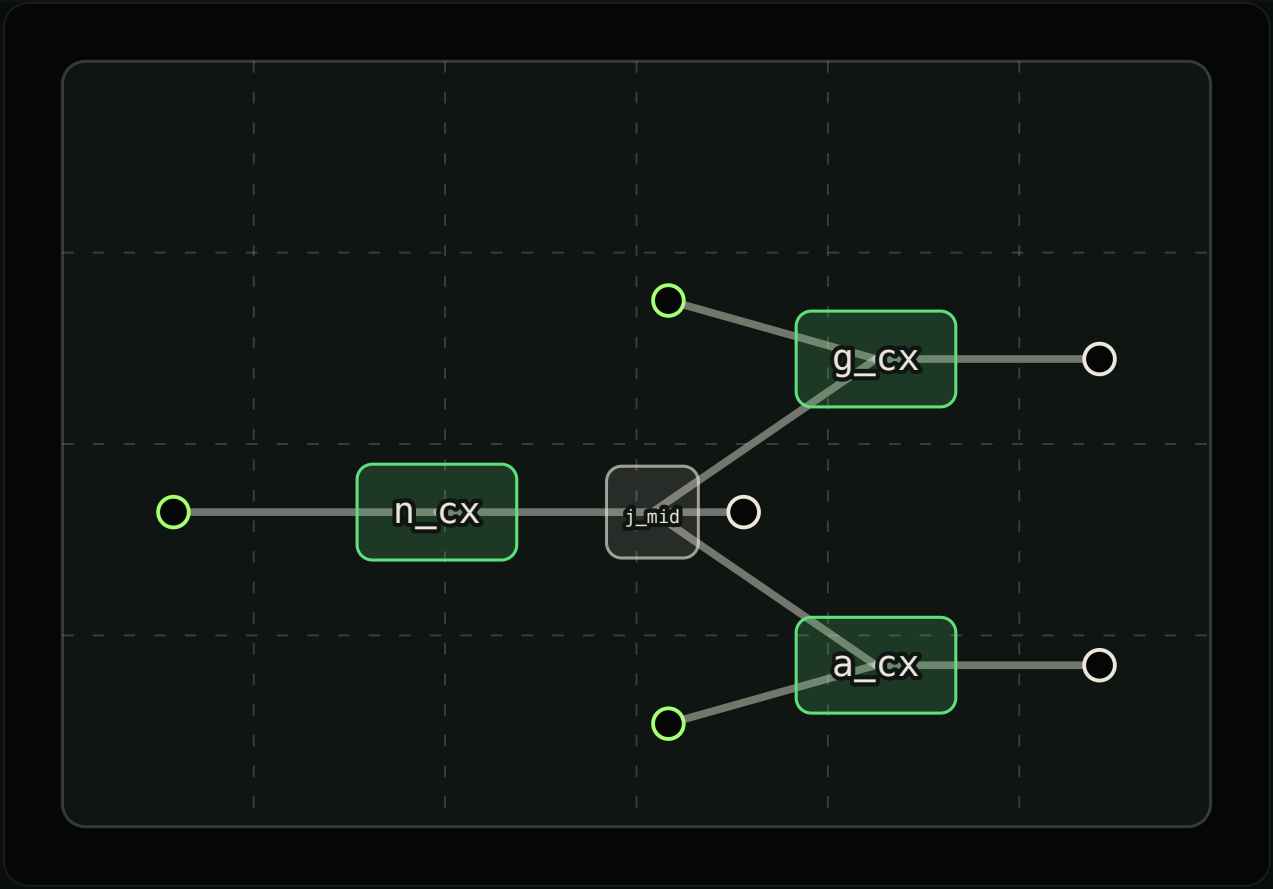
The bulk metabolome + proteome arm.

182	88 days	4	6
protocol steps	total protocol time	chambers	ports

0.3 mm
etch depth, one layer

THE CHIP, FROM ABOVE

Plan | Tilted



exchange pool channel: liquid port grid 5 mm

Drawn live from `chip/module1_v1.json`. One flat layer at one depth, so the view from above is the whole design. The tilted view exaggerates depth eight times.

PORTS

Holes drilled through the lid where tubing connects. Liquid goes in at the left, out at the right.



BENCH RECORDING

Recording not yet uploaded

Planned shot: Top-down macro of module1_v1 on a backlight, ruler in shot: food dye pushed in through its inlets, filling n_cx, g_cx, a_cx, j_mid in protocol order, then flushed out.

+ What this chip cannot do (6)

Recorded by its designers, so a demo chip is never mistaken for the real thing.

+ Full engineering detail

Every protocol step with its volumes and timings, the flow maps and the 3D simulation, straight from the chip spec.

← Bulk omics

Cell prep →

How to read the labels

Wherever a number could be mistaken for a fact it is not, it carries one of these.

SIMULATED

Simulated: a depth-averaged Hele-Shaw network solve, not a measurement from a physical chip. Shown in amber.

DEMO SCALE

Demo scale: 30 x 20 mm fabrication demo, not the ANSI/SLAS 127.76 x 85.48 mm design target.

PRINTED FOR THE DEMO

Printed for the demo: one of the six chips physically resin-printed for this weekend.

LLM STUB

LLM stub: extraction ran offline with the deterministic fallback, not a language model.

NOT YET IMPLEMENTED

Not yet implemented: this stage of the pipeline does not exist yet, so there is no output.

Words used on this site

Underlined words anywhere on the site show these definitions when you hover or focus them.

capsule

A hollow gel bead about 70 µm across, a little narrower than a hair, holding one cell. Its shell lets reagents in but keeps the cell's RNA inside, so liquids can be swapped around it and every cell stays separate.

lane

One of the three culture channels on Module 1, one per cell type: neurons, microglia and astrocytes. The lanes start out sealed from each other, are then connected so the cells can signal, and are harvested together at the end.

weir

A low step under the lid that leaves a gap narrower than a capsule. Liquid flows over it and capsules stay behind. We use height for this because our

shear

The drag that moving liquid puts on whatever it flows past, including cells on the floor. Too much

printers cannot make anything narrower than 0.2 mm side to side, but can control depth finely.

residence time

How long liquid stays in one spot before it is flushed out. A corner with a long residence time keeps old liquid in it, which means a wash or a reagent change did not fully happen there.

blocker

A problem severe enough that the protocol would not work on the chip as drawn, for example a pressure the chip cannot hold, or shear far above what the cells survive. High, medium and low are progressively milder.

reduced-order vs 3D

The fast flow simulation treats each shallow channel as a flat sheet (a "Hele-Shaw" model). That is the right physics for slow flow in flat channels and takes seconds. The 3D simulation (OpenFOAM) resolves the full depth and takes far longer; it is used to check the fast one where the flat-sheet idea is doubtful.

printed for the demo

One of the six chips physically resin-printed for this weekend's demonstration. The others exist as designs only.

LLM stub

The step that reads a protocol and pulls out its steps normally uses a language model. In stub mode it runs offline with simple fixed rules instead, so what it extracts is a placeholder, not a reading of the text.

and cells are damaged or washed off, so every step has a ceiling. Measured in dyn/cm².

sweep

Changing the liquid in a full chamber by pushing fresh liquid through it, usually a few chamber volumes, so the old liquid is carried out. Different from filling an empty chamber.

demo scale

The 30 × 20 mm chips we can print this weekend: a scaled-down test of the layout. The real device targets a standard microplate footprint, 127.76 × 85.48 mm. Numbers from a demo chip are not numbers for the real one.

off-chip

Done at the bench, in a tube, or on a bought instrument, not on a chip we print. Usually because the feature needed is smaller than our printers can make.

simulated

Worked out by the flow model from the chip geometry, not measured on a real chip. No pumps or cameras are connected this weekend, so every flow number here is a prediction.

handoff

Moving the sample from one chip to the next, usually through tubing. Each one costs some volume left in the tube (dead volume) and some pressure.

readout

One kind of measurement the assay produces. There are four: what the cells release (secretome), their small molecules (metabolome), their proteins (proteome) and, cell by cell, which genes are switched on (transcriptome).

barcode

A short unique DNA sequence attached to everything from one cell. After sequencing, reads with the same barcode came from the same cell, which is what makes the data single-cell.

trap (dead zone)

A region where liquid barely moves, so it is not exchanged when the rest of the chamber is. The flat-sheet model cannot show swirling eddies, so a trap here means slow flow or long residence time.

bubble

Air caught when a chamber fills, typically where a channel opens suddenly into something wider and the liquid front sticks at the edge while the corners fill last.

sediment

Cells or capsules sinking to the floor faster than the flow carries them, so they pile up instead of moving on.

leak

Liquid going somewhere it should not, for example out of a port meant to be closed, or into a neighbouring lane.

clog

A channel narrow enough that cells or capsules could jam in it.

NEXT

Run it

Give the software a protocol, or re-check these chips with the flow simulation.

OR

See results

The chips a run made, what the simulation found, and how to fix it.

ELSEWHERE

Digital twin

The same chips in 3D, animated through their protocol.

Runs entirely on this computer. No web fonts, no tracking, nothing fetched from the internet.

RUN IT

Turn a written lab protocol into chips, and check they flow.

Drop a protocol and the software works out which steps can run on a chip, designs and routes the chips, simulates the liquid moving through them and writes a report. It runs on this machine; nothing is uploaded anywhere else.

DESIGN FROM A PROTOCOL

Start from your protocol

About a minute for a short protocol. You can watch every stage as it happens.



Drop a protocol here

A PDF, markdown or plain-text file, or click to choose one

The file is saved under assay2chip/work/<run>/upload/ on this machine. The server only listens to this computer.

Design chips for this protocol

Paste text or a DOI instead

For a protocol that is not in a file.

Options

Chip size, how the protocol is read, and which stages to run. The defaults are fine.

CHECK WHAT EXISTS

Already have chips?

Re-check the six-chip chain shown on The assay page without redesigning anything.

This meshes every chip, solves the flow through each protocol step, joins the chips through their tubing and lists anything that would go wrong, such as too much shear, a long residence time, bubbles or leaks.

Simulate the existing chain ○ SIMULATED

No chip file is changed: a proposed fix is applied to a copy. Every number it produces is a prediction from the flow model.

Earlier runs

Pick one up where it left off, or open what it produced.

- `vof`

  08/11/2026, 12:35:13

  not recorded

  Progress Results

- `all_chips`

  08/11/2026, 12:33:41

  not recorded

Progress Results

- `foam`

  08/11/2026, 12:31:25

  not recorded

  Progress Results

- `foam_validate`

  08/11/2026, 12:29:41

  not recorded

  Progress Results

- `xval_coarse`

  08/11/2026, 12:03:50

  not recorded

  Progress Results

- `pipeline_mvp`

  08/11/2026, 11:27:01

   57 findings, 20 blocking

  Progress Results

- `pipeline_e2e`

  08/11/2026, 11:25:49

  not recorded

  Progress Results

- `xval`

  08/11/2026, 10:23:09

  not recorded

Progress Results

Runs entirely on this computer. No web fonts, no tracking, nothing fetched from the internet.

RESULTS

What one run produced, and what would go wrong.

The chips it designed, the flow simulation through them, every problem it found ranked by how serious it is with a proposed fix, its report, and the bench recordings.

SHOWING RUN

pipeline_mvp · 57 findings

NO STAGE RECORD LLM STUB DEMO SCALE SIMULATED Started 08/11/2026, 11:27:01

Stage progress

VERDICT CHIPS FLOW MAPS FINDINGS REPORT RECORDINGS EVERYTHING ELSE

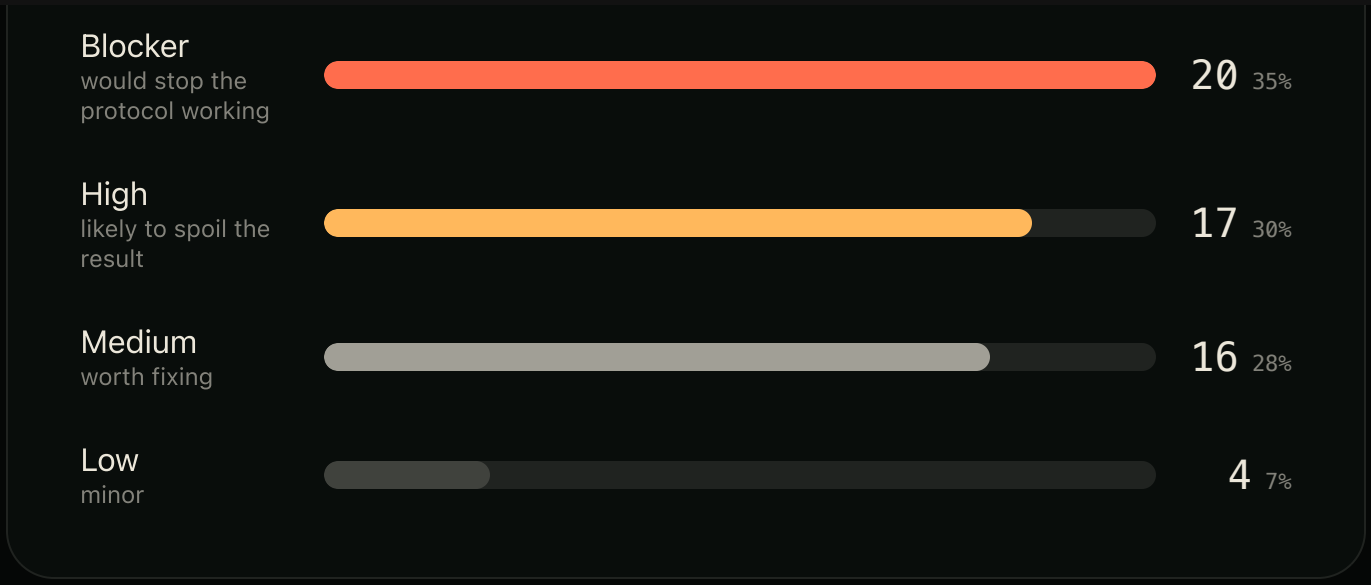
Will it flow?

The flow simulation's overall answer, and where the problems are.

No, not as drawn

20 blockers would stop the protocol working on these chips. Each comes with the two numbers that disagree and at least one fix.

SIMULATED WHOLE CHAIN



The flow model failed one of its own checks. Before trusting any number it gives, the solver runs itself against known answers. 8 passed and 1 failed, so every finding below is marked untrusted until that is fixed.

1 benchmark(s) FAILED: hele_shaw_validity (chambers below the aspect floor still meshed as a 2D field = 39). Every finding is untrusted until these pass.

WHERE THE PROBLEMS ARE

One square per finding, coloured by severity. Click a chip to list only its findings.

module1_v1	3 of 10 blocking
cell_prep_v1	2 of 9 blocking
spc_demo_v1	3 found
module3_v1_translated	2 of 8 blocking

module4_v1

8 of 17 blocking

module5_v1

5 of 10 blocking

127 fixes proposed. 4 applied to a copy of the chip and re-checked, 0 more could be applied automatically, 123 need a person to decide. The original chip files are never changed.

The chips it made

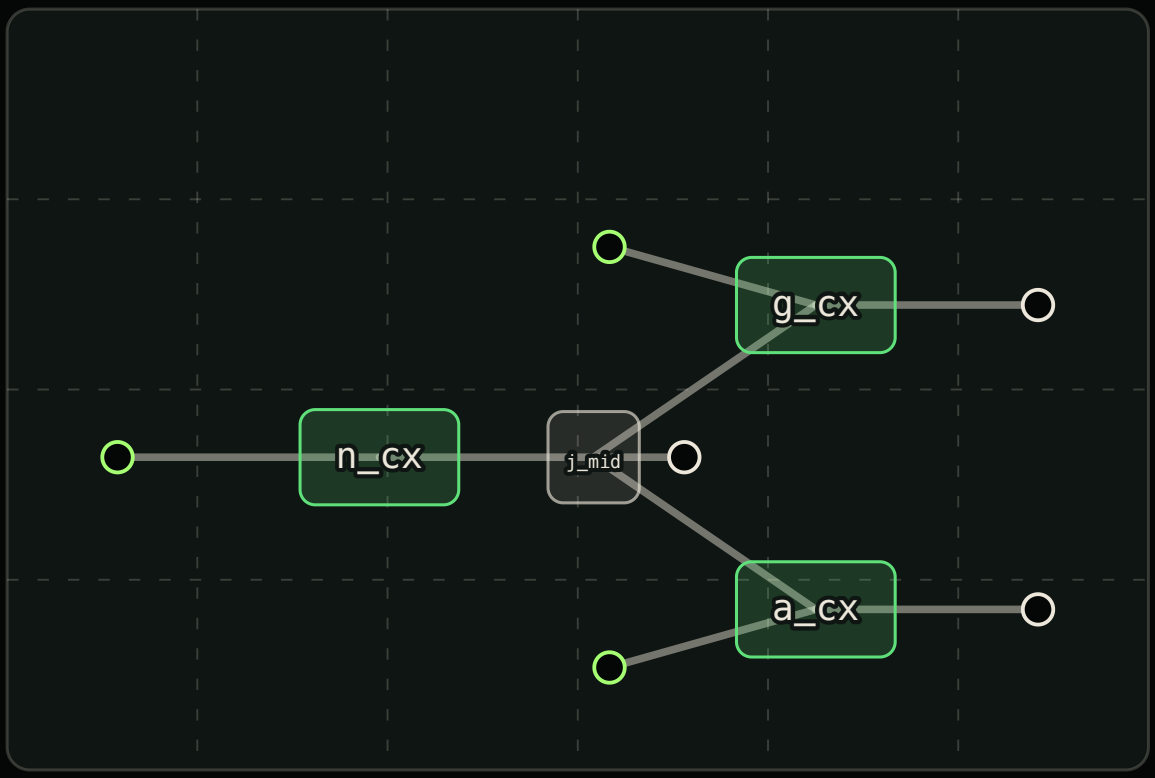
Each chip drawn from its design file. One flat layer at one depth, so the view from above is the whole design.

This run did not export printable 3D files, so each chip's tilted view is drawn from its design file instead. Depth is exaggerated so the 0.3 mm channels can be seen.

module1_v1

Module 1: flow-gated three-lineage chamber

PRINTED FOR THE DEMO · DEMO SCALE 30 × 20 mm



Plan | Tilted

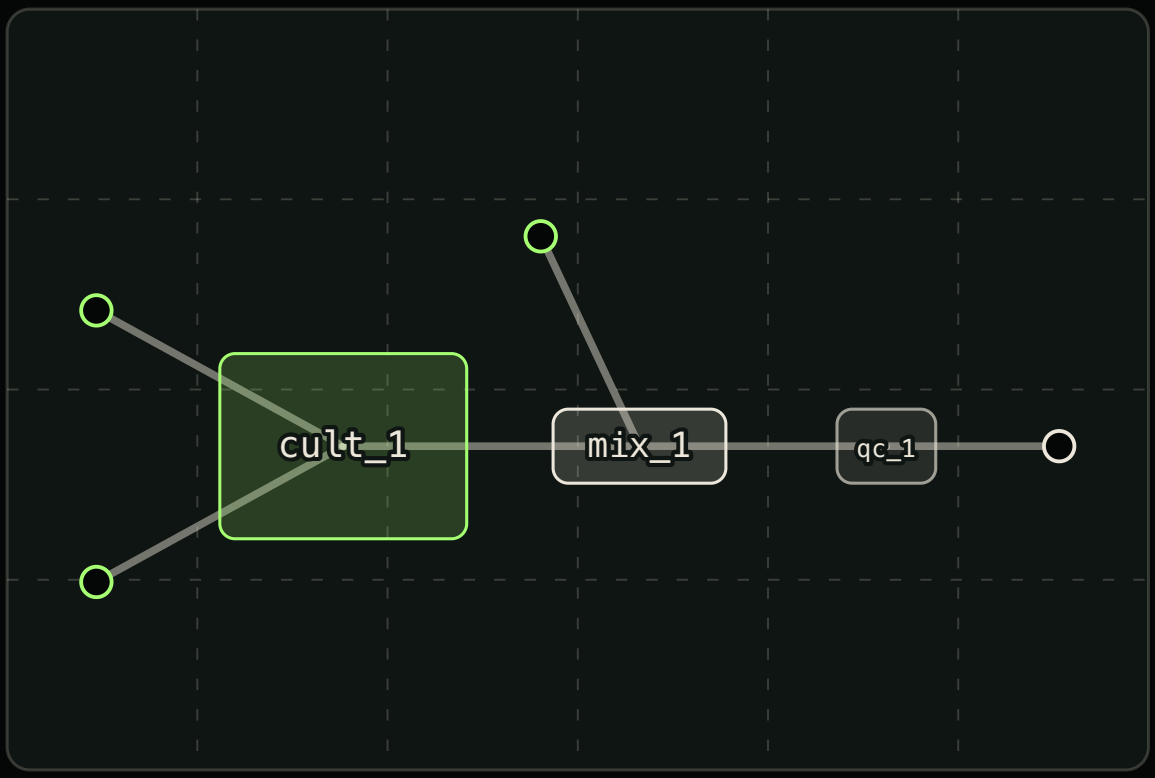
Flow maps →

10 findings, 3 blocking

cell_prep_v1

Cell detachment and QC module v1 (demo scale)

DEMO SCALE 30 × 20 mm



Plan | Tilted

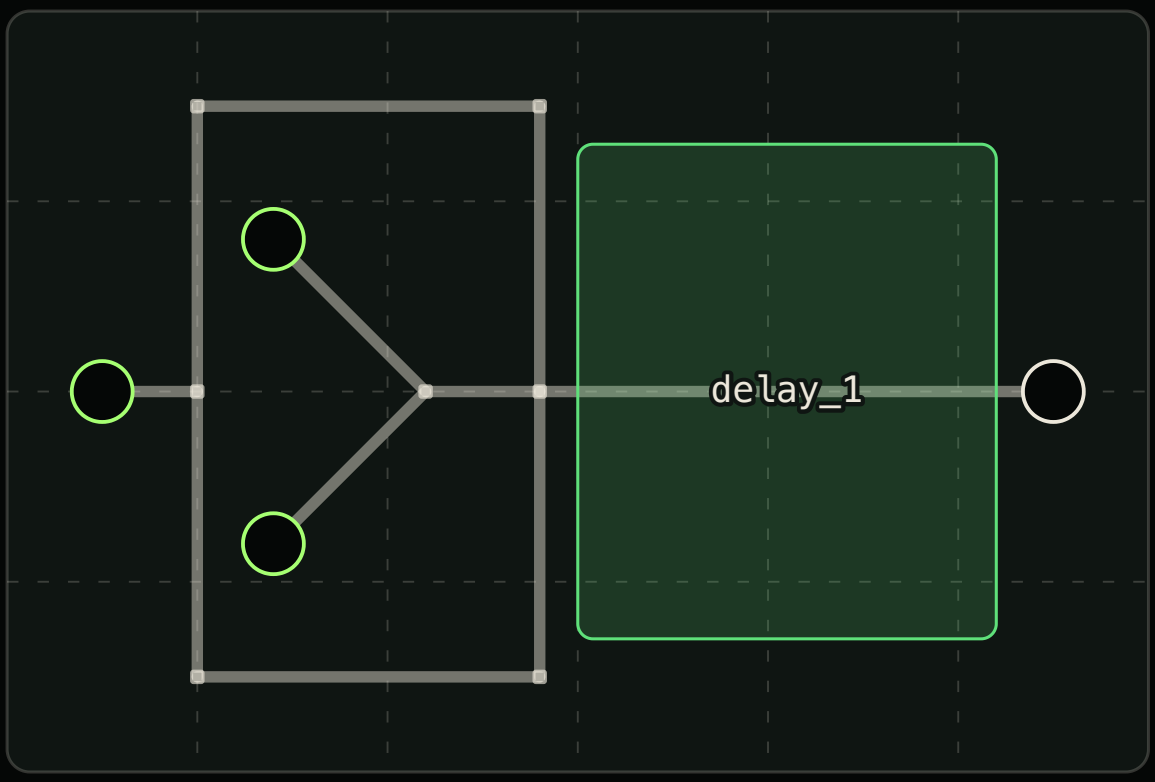
Flow maps →

9 findings, 2 blocking

spc_demo_v1

Flow-focusing geometry demo / reference board (not the pipeline's encapsulator)

PRINTED FOR THE DEMO | DEMO SCALE | 30 × 20 mm



Plan Tilted

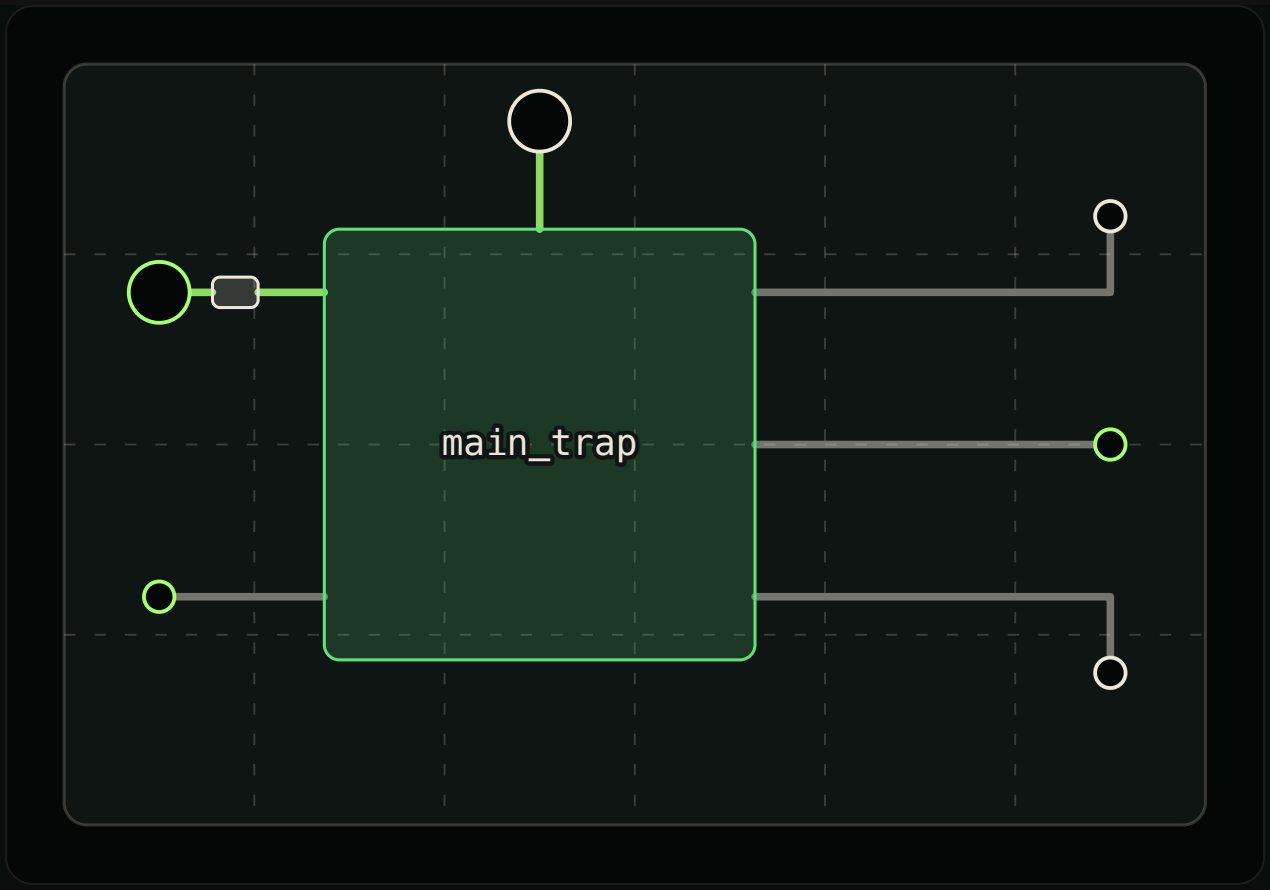
Flow maps →

3 findings

module3_v1_translated

Module 3: load, lysis, wash (translated from the project's geometry requirements sheet)

PRINTED FOR THE DEMO DEMO SCALE 30 × 20 mm



Plan | Tilted

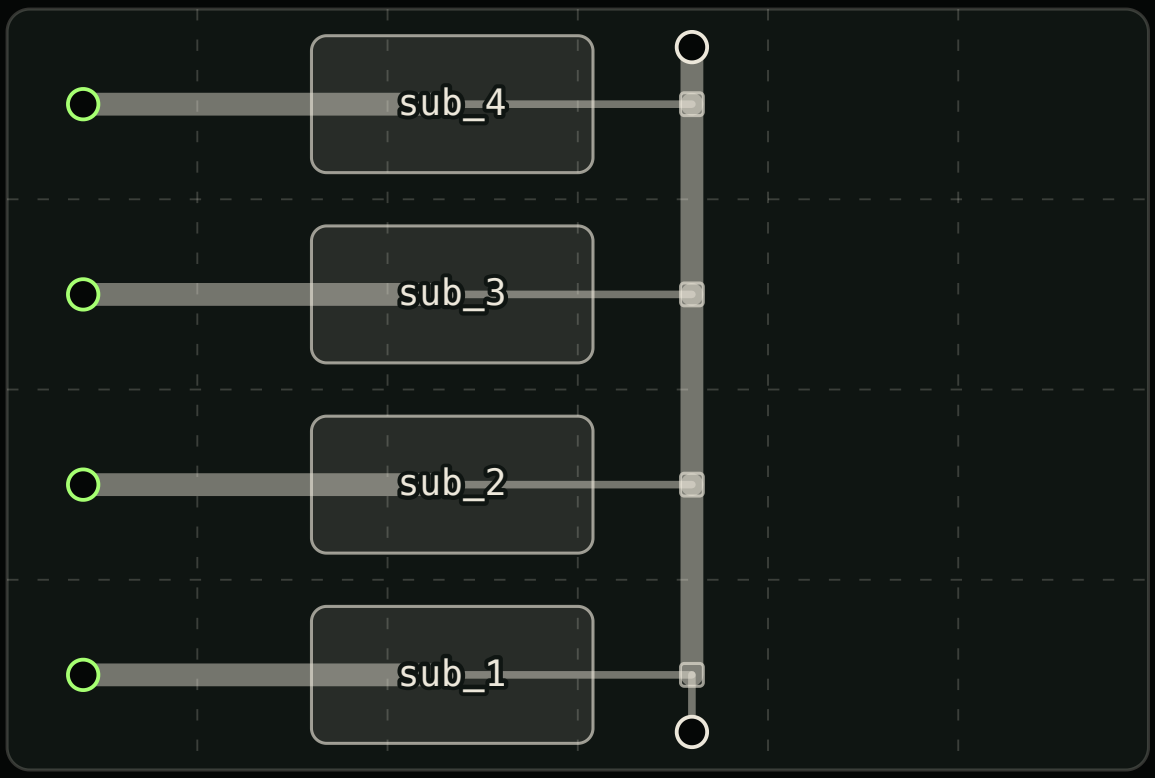
Flow maps →

8 findings, 2 blocking

module4_v1

Module 4: sublibrary split and index PCR

PRINTED FOR THE DEMO | DEMO SCALE 30 × 20 mm



Plan | Tilted

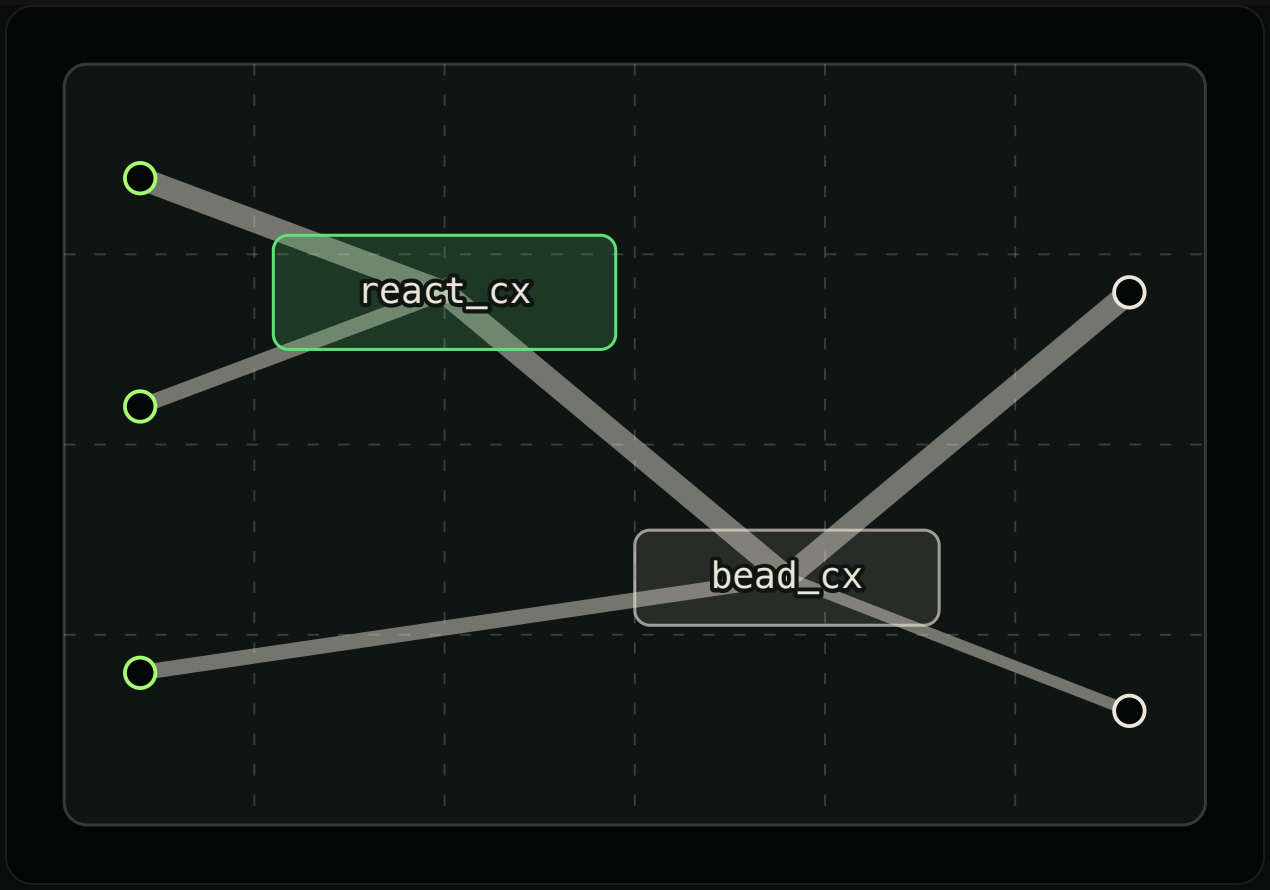
Flow maps →

17 findings, 8 blocking

module5_v1

Module 5: capsule release and library cleanup

PRINTED FOR THE DEMO | DEMO SCALE 30 × 20 mm



Plan | Tilted

Flow maps →

10 findings, 5 blocking

Flow simulation maps

What the liquid does inside one chip during one protocol step, coloured from low to high.

CHIP

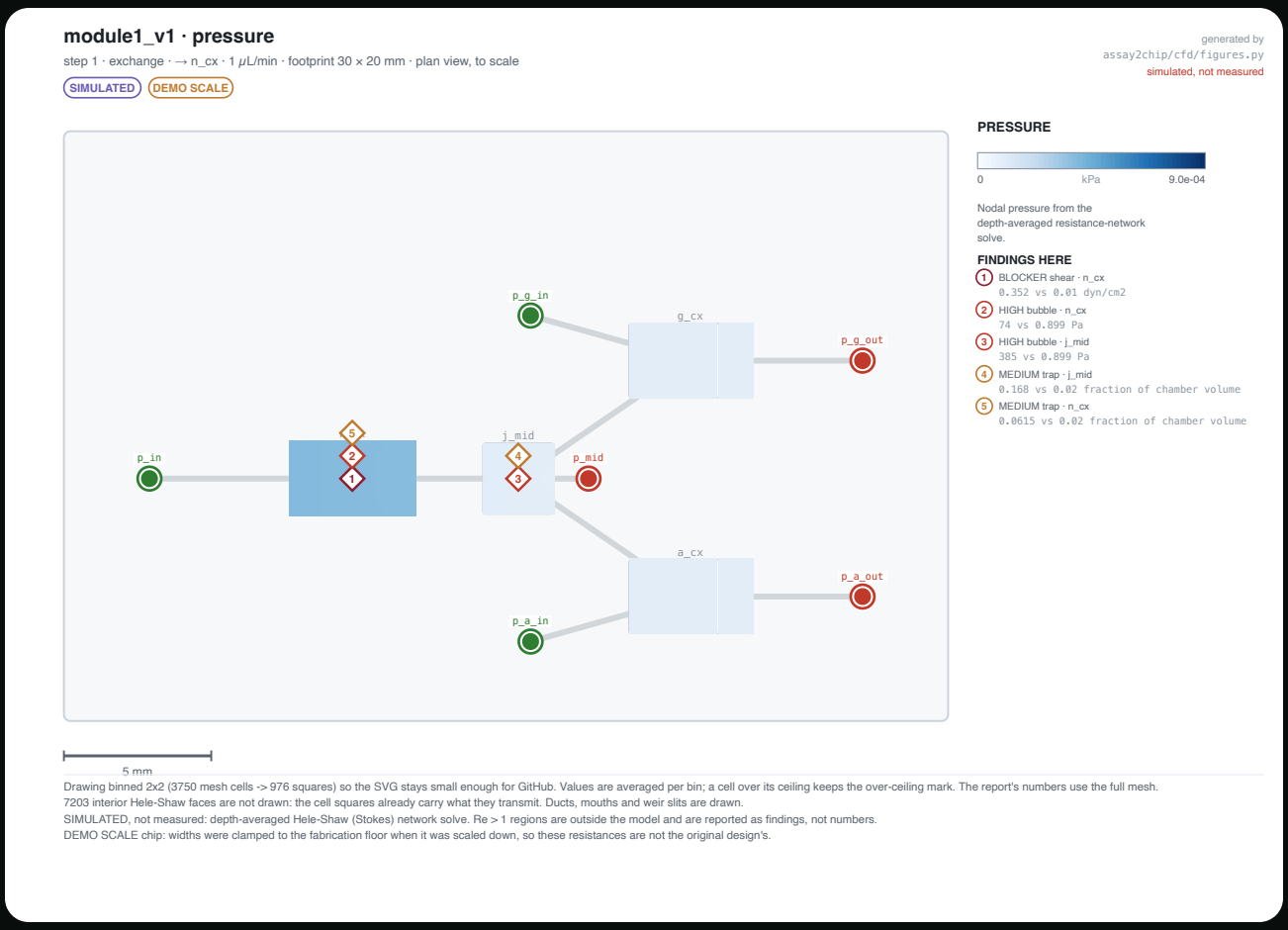
module1_v1 | cell_prep_v1 | spc_demo_v1 | module3_v1_translated | module4_v1 | module5_v1

PROTOCOL STEP

step 1 | step 3 | step 46 | step 48 | step 57 | step 46 (2)

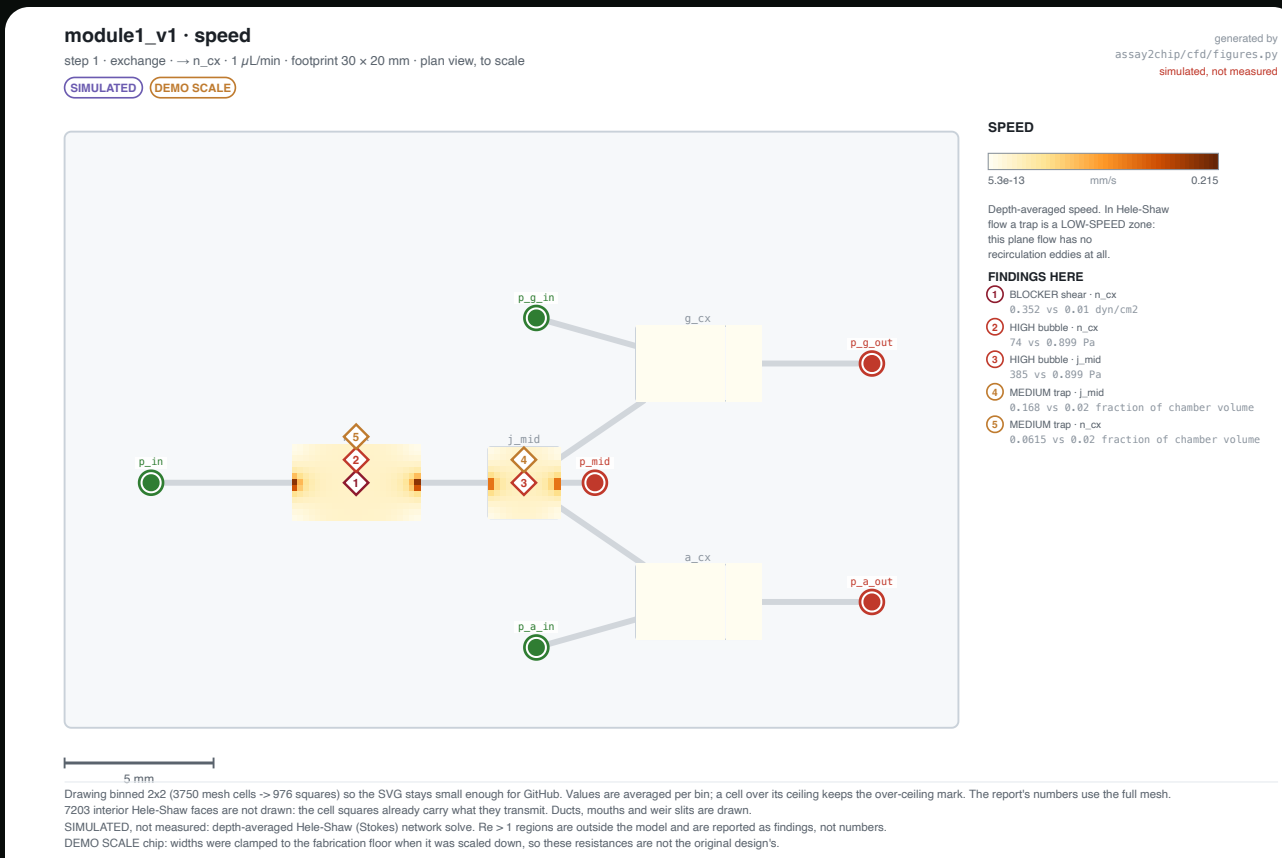
Step 1: exchange in n_cx at 1 µL/min INFERRED

From the fast flat-sheet model: see reduced-order vs 3D. Each map is simulated, not measured.



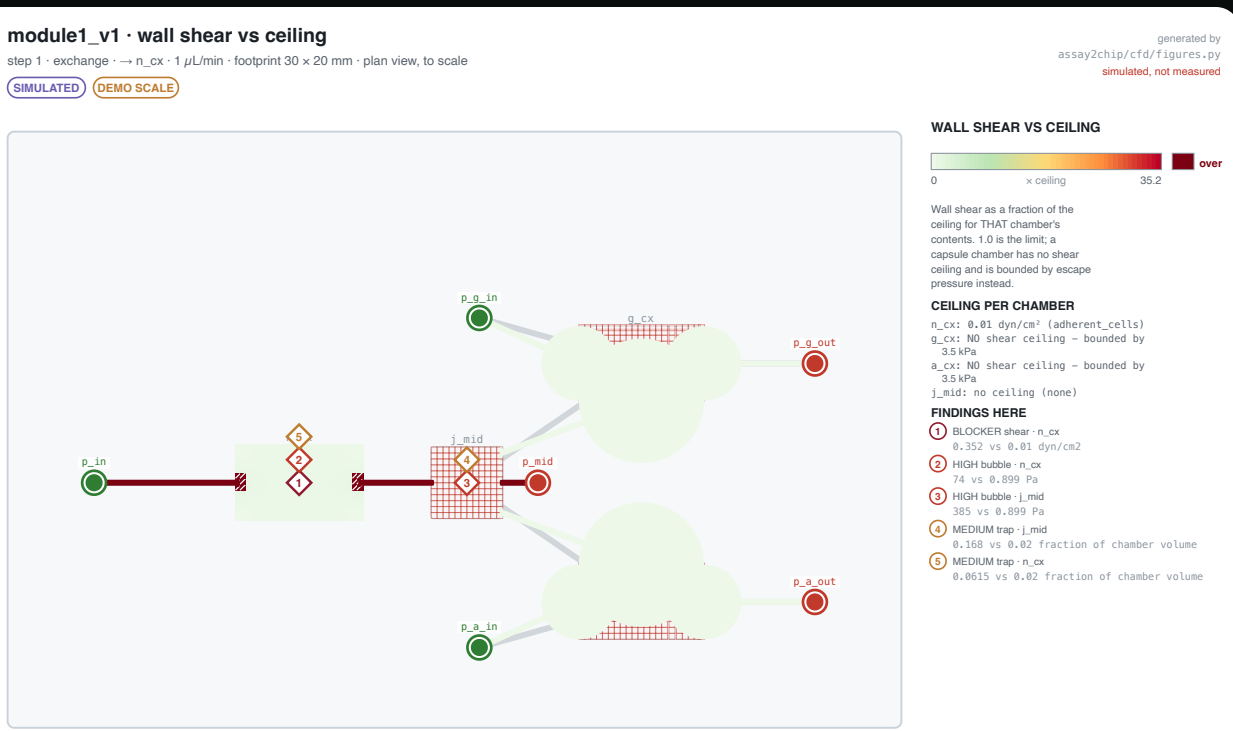
Pressure SIMULATED

How hard the pumps must push. A sudden jump marks a narrow restriction.



Speed ○ SIMULATED

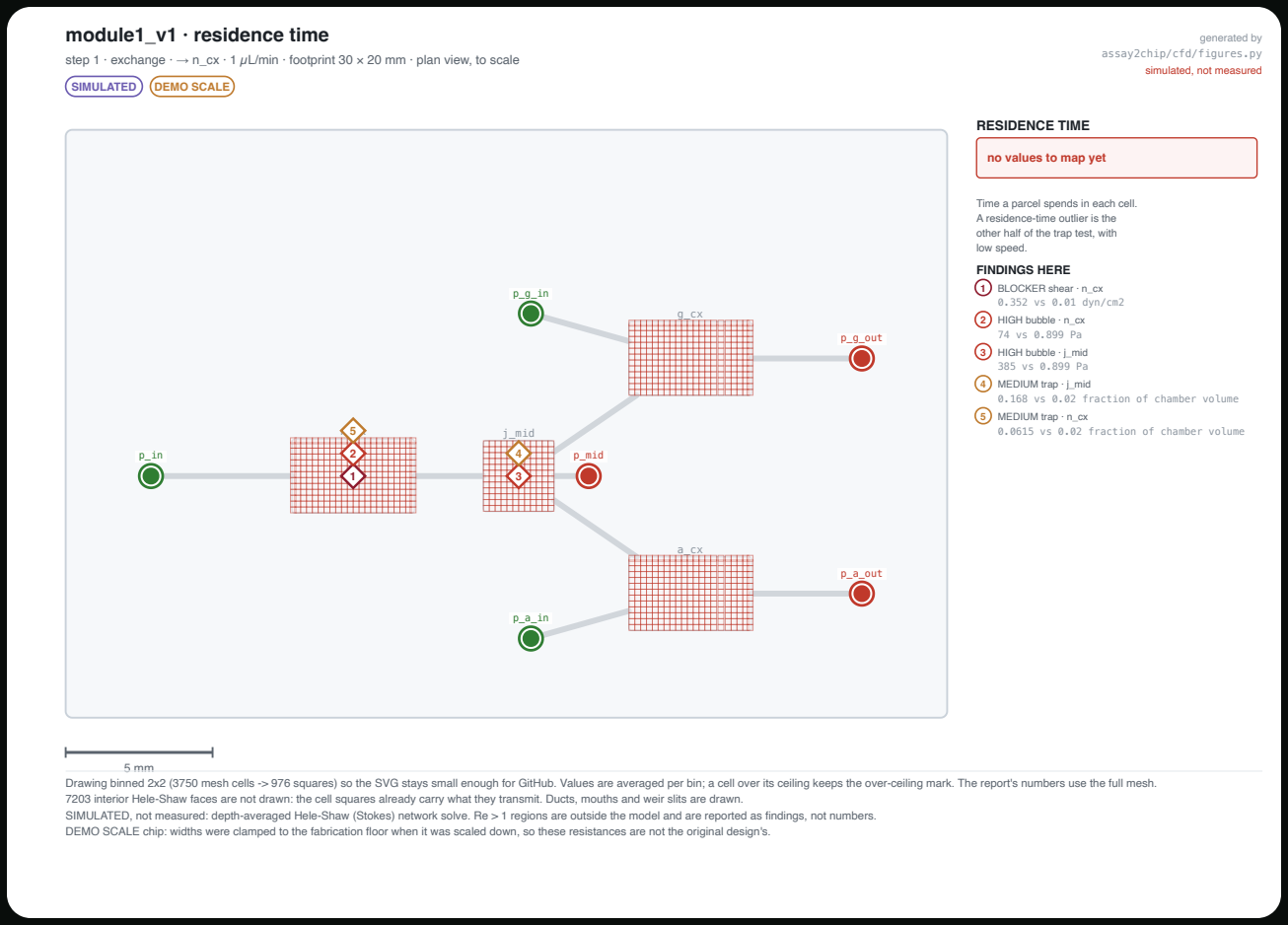
How fast the liquid moves. Slow, dark corners are where old liquid lingers.



Wall shear

SIMULATED

The drag on the floor where cells sit, against the ceiling for this step.



Residence time ○ SIMULATED

How long liquid stays in each spot. Long times mean a wash does not reach there.

Findings, most serious first

Every place the simulation says the chip and the protocol disagree, with the two numbers and a proposed fix.

All 57 | Blocker 20 | High 17 | Medium 16 | Low 4

Chip every chip

BLOCKER shear

Too much shear on sub_4

module4_v1 · step 4 UNTRUSTED



4.2 dyn/cm2 SIMULATED against 0.01 dyn/cm2 · 420× the limit

wall shear vs adherent cells ceiling

+ The simulation's own words

Fix: Change the flow rate (sub_4) 2.17 → 0.00797 dyn/cm2 SIMULATED

NEEDS A PERSON · 1 other option in the full tables below

BLOCKER shear

Too much shear on sub_3

module4_v1 · step 3 UNTRUSTED



4.2 dyn/cm2 SIMULATED against 0.01 dyn/cm2 · 420× the limit

wall shear vs adherent cells ceiling

+ The simulation's own words

Fix: Change the flow rate (sub_4) NEEDS A PERSON · 1 other option in the full tables below

---

BLOCKER shear

Too much shear on sub_2

module4_v1 · step 2 UNTRUSTED



4.2 dyn/cm2 SIMULATED against 0.01 dyn/cm2 · 420× the limit

wall shear vs adherent cells ceiling

+ The simulation's own words

Fix: Change the flow rate (sub_3) NEEDS A PERSON · 1 other option in the full tables below

---

BLOCKER shear

Too much shear on sub_1

module4_v1 · step 1 UNTRUSTED



4.2 dyn/cm2 SIMULATED against 0.01 dyn/cm2 · 420× the limit

wall shear vs adherent cells ceiling

+ The simulation's own words

Fix: Change the flow rate (sub_2) NEEDS A PERSON · 1 other option in the full tables below

---

BLOCKER shear

Too much shear on bead_cx

module5_v1 · step 1 UNTRUSTED



0.865 dyn/cm2 SIMULATED against 0.01 dyn/cm2 · 86× the limit

wall shear vs adherent cells ceiling

+ The simulation's own words

Fix: Change the flow rate (bead_cx) 0.435 → 0.008 dyn/cm2 SIMULATED NEEDS A PERSON · 1 other option in the full tables below

---

BLOCKER sediment

Particles could settle in ch_in_bead

module5_v1 · step 1 UNTRUSTED



79.4 transit/settling SIMULATED against 1 transit/settling · 79× the limit

transit time over settling time

+ The simulation's own words

Fix: Increase the flow rate (ch_in_bead) NEEDS A PERSON · 2 other options in the full tables below

---

BLOCKER sediment

Particles could settle in ch_react_bead

module5_v1 · step 1 UNTRUSTED



74.4 transit/settling ○ SIMULATED against 1 transit/settling · 74× the limit

transit time over settling time

+ The simulation's own words

Fix: Increase the flow rate (ch_react_bead) NEEDS A PERSON · 2 other options in the full tables below

---

BLOCKER shear

Too much shear on a_cx

module1_v1 · step 57 UNTRUSTED



0.515 dyn/cm2 ○ SIMULATED against 0.01 dyn/cm2 · 51× the limit

wall shear vs adherent cells ceiling

+ The simulation's own words

Fix: Change the flow rate (a_cx) 0.515 → 0.00798 dyn/cm2 ○ SIMULATED APPLIED TO A COPY, RE-CHECKED

BLOCKER shear

Too much shear on g_cx

module1_v1 · step 46 UNTRUSTED



0.515 dyn/cm2 SIMULATED against 0.01 dyn/cm2 · 51× the limit

wall shear vs adherent cells ceiling

+ The simulation's own words

Fix: Change the flow rate (g_cx) 0.515 → 0.00798 dyn/cm2 SIMULATED

APPLIED TO A COPY, RE-CHECKED

BLOCKER sediment

Particles could settle in ch_in_lib

module5_v1 · step 1 UNTRUSTED



38.6 transit/settling SIMULATED against 1 transit/settling · 39× the limit

transit time over settling time

+ The simulation's own words

Fix: Increase the flow rate (ch_in_lib) NEEDS A PERSON · 2 other options in the full tables below

Show all 57

Report

The run's own written summary: problems, fixes and the files it made.

SIMULATED

S10 CFD: pipeline_mvp

MVP pipeline (capseq): CapSeq (RNA): culture -> cell prep -> encapsulation -> load, lysis, wash, reverse transcription (one trap) -> sublibrary PCR -> release and cleanup -> SPRI and QC (off-chip) (+ barcoding and hairpin placeholders)

> ## VALIDATION FAILED - ALL FINDINGS BELOW ARE UNTRUSTED
>
> 1 benchmark(s) FAILED: hele_shaw_validity (chambers below the aspect floor still meshed as a 2D field = 39). Every finding is untrusted until these pass.. Until the benchmarks in `cfd/README.md` pass (analytic duct resistance, series and parallel networks, agreement with `operator/network.py`, mass conservation) every number in this report is the output of an unchecked solver. Read the findings as *where to*

Read the whole report

Download

Bench recordings

One slot per printed chip: a real recording of dyed water run through it, once filmed.

MODULE 1

module1_v1

● PRINTED FOR THE DEMO

IN THIS RUN

Not recorded yet

Planned: Top-down macro of module1_v1 on a backlight,

MODULE 2

spc_demo_v1

● PRINTED FOR THE DEMO

IN THIS RUN

Not recorded yet

Planned: Top-down macro of spc_demo_v1 on a backlight,

ruler in shot: food dye pushed in through its inlets, filling n_cx, g_cx, a_cx, j_mid in protocol order, then flushed out.

ruler in shot: food dye pushed in through in_oil, in_shell, in_core, filling delay_1, xj_1 in protocol order, then flushed out.

MODULE 3

module3_v1_translated

● PRINTED FOR THE DEMO

IN THIS RUN

Not recorded yet

Planned: Top-down macro of module3_v1_translated on a backlight, ruler in shot: food dye pushed in through capsule_in, reagent_in, backflush_in, filling main_trap in protocol order, then flushed out.

MODULE 4

module4_v1

● PRINTED FOR THE DEMO

IN THIS RUN

Not recorded yet

Planned: Top-down macro of module4_v1 on a backlight, ruler in shot: food dye pushed in through in_s1, in_s2, in_s3, in_s4, filling sub_1, sub_2, sub_3, sub_4 in protocol order, then flushed out.

MODULE 5

module5_v1

● PRINTED FOR THE DEMO

IN THIS RUN

Not recorded yet

Planned: Top-down macro of module5_v1 on a backlight, ruler in shot: food dye pushed in through in_lib, in_bead, in_prot, filling react_cx, bead_cx in protocol order, then flushed out.

MODULE 6

module6_v1

● PRINTED FOR THE DEMO

Not recorded yet

Planned: Top-down macro of module6_v1 on a backlight, ruler in shot: food dye pushed in through in_n, in_g, in_a, filling cx_n, cx_g, cx_a in protocol order, then flushed out.

Everything else

The full detail behind this page, for when you need the exact files and numbers.

The stage-by-stage record

Every pipeline stage with its timing, the files it wrote (opened in place) and the full log.

Where each protocol step ended up

Every step read from the protocol and whether it landed on a chip, off-chip, or needs a part that does not exist.

Full flow-simulation tables

Every finding and every fix as a table, the solver's own checks, the tubing between chips and which parts of the solver ran.

Every chip in the repo

All chip designs, not only this run's: plan, 3D, fabrication limits and downloads.

The report, full page

The report with a table of contents.

Runs entirely on this computer. No web fonts, no tracking, nothing fetched from the internet.